

# Nwele Uchenna David

developeruche@gmail.com

Systems Engineer - Rust, Ethereum Execution Layer, zkVMs

Open to: Remote (global), Relocation

developeruche.com

github.com/developeruche

linkedin.com/in/developeruche

## Work Experience

### • Ethereum Foundation - zkEVM Team

Remote

*Research Engineer (Contract)*

*May - August 2026*

- Proposed and drafted a REST/SSZ `newPayloadWithWitness` Engine API endpoint for Reth and Ethrex, motivated by EIP-8025 and real-time zkVM attestation; benchmarks showed order-of-magnitude witness-delivery latency reduction over the existing round-trip.
- Implemented `zilkworm-stateless`, a C++ zkVM guest program for stateless Ethereum block validation, covering EEST spec compliance and Merkle-Patricia Trie changes.
- Contributed to `ere-guests`, a collection of stateless block-validation guest programs for Ethereum blocks, and to `ere`, the Foundation's unified zkVM interface and toolkit.
- Integrated a purpose-built TCP engine into Scroll's `reth-witness-indexer`.

### • Ethereum Foundation - zkEVM Team

Remote, Buenos Aires, Argentina

*Research Engineer, Ethereum Protocol Fellowship*

*June - Nov 2025*

- Benchmarked precompiles across four major zkVM platforms (RISC0, SP1, OpenVM, Ziren), showing that optimized software with lean precompiles outperforms generic implementations with fat precompiles by roughly 27×; published the findings as *Fat vs. Lean Precompiles in zkVMs*, arguing for a precompile strategy that avoids ecosystem fragmentation.
- Identified AOT compilation as the primary lever for 10× zkVM performance gains, reaching 36M gas in 0.5s on consumer hardware, after profiling trace-generation bottlenecks.
- Designed and prototyped a hybrid Ethereum architecture replacing the EVM with a RISC-V execution engine while preserving backward compatibility, via magic-number detection and a mini-EVM interpreter running as a RISC-V program.
- Integrated the hybrid VM into a production Reth node and built `cargo-hybrid`, a CLI toolchain for scaffolding, compiling and deploying RISC-V smart contracts; benchmarked end-to-end against `revm`.
- Prototyped "Stateless Attestors" with the Prysm team, enabling consensus-layer nodes to verify execution payloads through zkEVM proofs without a full execution-layer dependency.
- Built a stateless Ethereum block validator running inside `ziskemu`; compiled Geth bare-metal with the Tamago toolchain to eliminate Linux syscall dependencies on zkVM targets.

### • Independent Research - Rust Systems, zkVMs & ZKPs

Lagos, Nigeria

*Research Engineer (self-directed)*

*June 2024 - June 2025*

- Stepped back from full-time engineering to work on cryptography and zero-knowledge full time
  - reading papers daily, running research calls with peers, and implementing each result in Rust rather than stopping at the write-up; converged on zkVMs as the specialism.
- Built the polynomial layer the rest of the work rests on - univariate and multilinear representations with their evaluation, interpolation and arithmetic.
- Implemented the sum-check protocol and GKR, then succinct GKR - GKR composed with a multilinear KZG commitment scheme.
- Implemented Groth16, and KZG polynomial commitments in both univariate and multilinear form.
- Implemented PLONK end to end - prover, verifier, circuit compiler, and a transpiler lowering GKR circuits to Plonkish arithmetization.
- Published the whole body of work as `cryptography-n-zk-research`, an open repository with accompanying notes on computational proof systems.

### • MyAI - Me Protocol

USA, Remote

*Senior Blockchain Engineer*

*Nov 2022 - July 2025*

- Led the blockchain engineering team delivering Me Protocol V3, V4 and V5 alongside the CTO, owning architecture through to mainnet deployment.
- Architected the protocol in Solidity and Yul on a custom Diamond-standard design, with Foundry-based deployment and testing and an in-house migration toolkit supporting both upgrades and

rollbacks across a system of contracts.

- Led development of a custom optimistic rollup in Rust settling to Polygon for asset transfer and L2-to-L1 bridging; omitting a sandboxed VM kept it substantially lighter than a general-purpose EVM rollup for the workload.

- **Polytope Labs - Hyperbridge**

*Blockchain Engineer*

Remote

*Jan - April 2024*

- Rust engineer on Tesseract, the Hyperbridge cross-chain relayer, and Substrate runtime engineer on the Nexus runtime deployed to Polkadot mainnet.
- Built HyperClient, a Rust library compiled to WebAssembly that lets Hyperbridge clients manage in-flight ISMP requests; implemented teleport support for DOT addresses.

- **Scroll**

*Smart Contract Engineer (Contract)*

Remote

*April - June 2024*

- Built the on-chain contract layer for Scroll Pass, a zero-knowledge event ticketing platform delivered with the Ethereum Community Fund.

- **Nethermind**

*Smart Contract Engineer (Intern)*

Remote

*Jan - April 2023*

- **Valantra:** built a DEX combining a Limit Order Book with constant-product and constant-sum AMM algorithms over a shared liquidity base, avoiding fragmentation; exchange, pool and management logic fully modularized.
- **Libre:** contributed to a decentralized stock-market protocol for on-chain IPOs.

- **AfriOne, Nahmii & Web3Bridge**

*Smart Contract Engineer, Design Architect & Team Lead (Contract)*

Remote / Lagos, Nigeria

*2021 - 2023*

- **AfriOne:** led the blockchain team; shipped a contract system on a private Geth network backing a consumer finance product, plus an SDK for the mobile and backend teams.
- **Nahmii:** built Bondii, an asset-bonding system, and a cross-chain staking platform unifying assets across Ethereum L1 and Nahmii 3; ported the Uniswap contracts and UI to Nahmii 3.
- **Web3Bridge:** smart contract team lead for an on-chain governance system built on the Diamond standard.

## Open Source & Research

- **Ethereum core tooling in Rust - Alloy, REVM, OpenZeppelin Stylus** *Merged contributions*

- Contributor to **Alloy** (the ground-up rewrite of ethers-rs), **REVM** (the Rust EVM used across the client ecosystem), and **OpenZeppelin's** Stylus contract library.

- **RISC-V EVM Experiment - research & Rust proof of concept**

*2025*

- Evaluated RISC-V as an alternative execution environment for smart contracts; published *Bridging Worlds: A Performance and Feasibility Analysis of RISC-V Integration with the Ethereum Virtual Machine*. Shipped a supporting RV32IM assembler and emulator in Rust.

## Skills

- **Languages:** Rust, Solidity, C++, TypeScript, Go
- **Ethereum:** Reth, REVM, Alloy, Foundry, Engine API, SSZ, Merkle-Patricia Tries, EIP authorship
- **Zero-Knowledge:** zkVMs (RISC0, SP1, OpenVM, Ziren, ZisK), Groth16, PLONK, KZG, GKR, sum-check
- **Systems:** RISC-V (RV32IM), compilers, emulators, AOT compilation, performance benchmarking, WebAssembly
- **Distributed Systems:** Substrate / Polkadot SDK, optimistic rollups, cross-chain messaging (ISMP)

## Education

- **University of Benin - B.Eng, Computer Engineering** *Benin City, Nigeria* *2019 - 2024*

- **Polkadot Blockchain Academy**

*Blockchain Development - Rust, Substrate Runtime Engineering*

Singapore

*May - June 2024*